Nuclear Plant Public Alert System

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Software Testing and Validation (ENSE 375)

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Agenda

- Introduction
- Problem Definition
- Design Requirements
- Solutions
- Testing and Demonstration
- Project Management
- Conclusion and Future Scope

Introduction

Background Information:

- Accidental Pickering, Ont Nuclear Alert
- Nuclear power plants require reliable emergency alert systems
- False alarms can cause panic and reduce public trust
- Operators must be able to test the system safely
- Human error is a major risk during training exercises
- A structured alert system helps prevent accidental live alerts

Rationale:

- A Public Alert System is needed to send accurate emergency notifications
- The system must reduce human error and prevent false alarms
- Layered verification improves safety and reliability
- Ensures alerts are sent only during real emergencies

Problem Definition

Public alert system must provide concise information and relevant instructions for a given emergency, and reduce the chance of human error triggering false alarms by having multiple conditions required to transition from training mode to live. This allows for safe testing without causing public panic and alarm fatigue, therefore real emergency alerts can be reacted to in an appropriate way.

- Public alert systems must be reliable and accurate
- Multiple levels of verification can lead to fewer accidental emergency notifications
- System needs safe testing without notifying the public

Design Requirements

Functions:

- Send alerts to the Emergency Planning Zone (EPZ)
- Allow training without triggering real alerts
- Log alert activity for auditing

Objectives:

- Minimize human error
- Prevent false alarms
- Ensure fast and reliable alerts
- Provide safe operator training

Constraints:

- Only Authorized users can send alerts
- System must be reliable at all times
- Must be cost-effective
- Avoid unnecessary public panic

Solutions

Feature	Solution 1	Solution 2	Final Solution
Architecture	Dashboard-centered with loosely defined classes	Improved structure with testing ideas	Fully Modular PAS with clear components Observer pattern to send alerts to people within the EPZ as well as trigger other alert methods like sirens and digital signs. Decorator pattern is used to form the emergency messages that are sent to people.
Seperation of Components	Limited separation	Partial separation	Clear separation (sensors, processing, state, alerts)
Boundary Value Analysis	Not clearly supported	Identified but not fully structured	Fully supported in sensor modules
Equivalence Class Testing	Not defined	Partially defined	Clearly defined inout ranges
Decision Table Testing	Difficult to apply	Proposed but loosely connected	Applied to processing and alert logic
State Transition Testing	No defined state model	Finite State Machine proposed	State Management Engine implemented
Testability	Low	Moderate	High
Input Validation	Not clearly handled	Mentioned but not fully structured	Implemented in sensor classification
Modularity	Low	Medium	High
Reason for Rejection/Selection	Hard to test	Better but not fully modular	Fully Best support for systematic testing

Testing and Demonstration

For the final solution, the following testing techniques were used:

- Boundary Value Analysis
- Equivalence Class Testing
- Decision Table Testing
- State Transition Testing
- Functional Testing

The test cases covered the following components:

- Temperature module
- Pressure module
- Radiation module
- Seismometer module
- DPU decision table
- State engine
- Alert system

Testing Results:

- Sensor modules correctly classified readings into expected alert levels
- Boundary values were handled correctly
- Decision table logic worked as expected using the “highest severity wins” rule
- State transitions behaved correctly, including latch behaviour in the Critical state
- Alert system produced the correct output for each system condition
- All test cases passed successfully when executed in the terminal

To show the progression of our work on this project, we created a Gantt chart that outlines key tasks, milestones, and the timeline for completing each phase.

Conclusion and Future Work

Overall, for this project we:

- Developed a modular Public Alert System (PAS) prototype for nuclear plant monitoring
- Implemented sensor modules for temperature, pressure, radiation, and seismic activity
- Applied boundary value analysis, equivalence class testing, decision tables, and state transition testing
- Verified correct alert classification and system state behaviour
- Met design objectives such as reducing human error and preventing false alarms

Future Work(if continued):

- Integrate real hardware sensors instead of manual input
- Add redundancy and fault-tolerance for safety-critical operation
- Expand monitoring to include more reactor parameters
- Improve real-time communication and alert delivery
- Implement advanced thresholds based on industry standards

https://youtu.be/VAZ8wE3Zzek